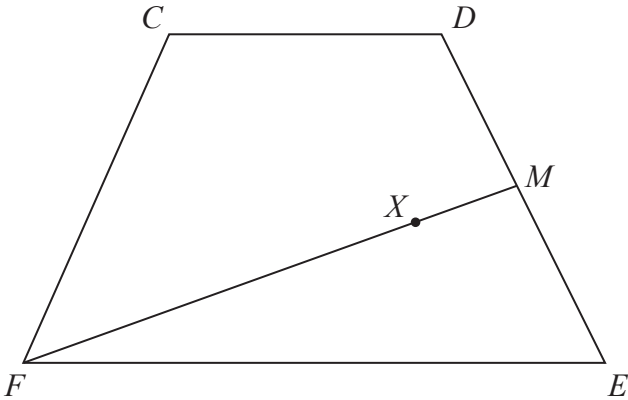


20 *CDEF* is a quadrilateral.



$\overrightarrow{CD} = \mathbf{a}$, $\overrightarrow{DE} = \mathbf{b}$ and $\overrightarrow{FC} = \mathbf{a} - \mathbf{b}$.

- (a) Express $\overrightarrow{FE}$ in terms of $\mathbf{a}$ and/or $\mathbf{b}$.
Give your answer in its simplest form.

(2)

M is the midpoint of *DE*.
X is the point on *FM* such that $FX:XM = n:1$
CXE is a straight line.

- (b) Work out the value of *n*.

$n =$
(4)